Q1. ALU is an integrated part of Bus Interface Unit (BIU). Do you agree? State your statement.

Answer:
No, I do not agree.

In the 8086 microprocessor, the **ALU (Arithmetic Logic Unit)** is a part of the **Execution Unit (EU)**, not the **Bus Interface Unit (BIU)**.

ALU performs arithmetic and logical operations such as addition, subtraction, AND, OR, etc.

Q2. What's the purpose of PC in an 8086 microprocessor?

Answer:
PC (Program Counter), known as **Instruction Pointer (IP)** in 8086, stores the address of the next instruction to be executed.

Functions:

- Keeps track of program execution.
- Points to the next instruction in memory.
- Updates automatically after each instruction execution.

Q3. How does 8086 differentiate a memory and I/O operation?

Answer:
The 8086 uses the **M/IŌ (Memory/Input-Output)** signal to distinguish between memory and I/O operations.

- **M/IO = 1** → Memory operation
- **M/IO = 0** → I/O operation

Thus, the processor can identify whether it is communicating with memory or an I/O device.

Q4. Show with an example how a 20-bit physical address is made in 8086 stack operation.

Answer:
In 8086, the physical address is calculated as:

Physical Address = Segment Address × 10H + Offset Address

Example:

SS = 2000H

SP = 3000H

Physical Address

$= 2000H \times 10H + 3000H$

$= 20000H + 3000H$

= **23000H**

Therefore, the 20-bit physical address is **23000H**.

Q5. How does 8086 react if any unusual operation occurs like overflow, carry or division by zero?

Answer:
When an unusual condition occurs, the 8086 updates the **Flag Register** or generates an interrupt.

- Overflow → Overflow Flag (OF) is set.
- Carry → Carry Flag (CF) is set.
- Divide by zero → Type-0 Interrupt is generated.

Thus, the processor detects and handles errors through flags and interrupts.

Q6. During an arithmetic operation, you need to generate an output at port X. How shall 8086 react and which pins shall be used?

Answer:
To send data to Port X, the 8086 executes an **OUT instruction**.

Example:

```
MOV AL, 25H
OUT X, AL
```

Pins used:

- Address Bus (A0–A15)
- Data Bus (D0–D15)
- $M/\overline{IO}$ pin
- $W\overline{R}$ (Write) pin

The processor places the port address on the address bus and sends data through the data bus to the output port.